

Notes: Yi (JPE, 2003)
Can Vertical Specialization Explain the Growth of World Trade?

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1 One-sentence summary

This note summarizes Yi (2003), which uses U.S. and world manufacturing export shares, tariff data, input-output based measures of vertical specialization, and a calibrated two-country dynamic Ricardian model with sequential production to ask why trade grew so much from modest tariff reductions and why the tariff-trade relationship became nonlinear after the mid-1980s; the paper concludes that vertical specialization magnifies tariff reductions through repeated border crossings and can explain more than half of U.S. manufacturing export growth, much of the growth of vertically specialized trade, and larger welfare gains than a standard one-stage trade model.

2 Big picture

- **Question.** Can falling tariffs explain the postwar growth of world trade once production is fragmented across countries rather than completed entirely inside one country?
- **Puzzle 1: magnitude.** Since the early 1960s, manufacturing tariffs fell by only about 11 percentage points, but the world manufacturing export share of GDP rose by a factor of about 3.4. Standard trade models need implausibly high substitution elasticities to match this.
- **Puzzle 2: nonlinearity.** Most tariff reductions happened before the mid-1980s, but trade responded much more strongly after the mid-1980s. Standard models typically predict a roughly linear or even weakening trade response as tariffs decline.
- **Core mechanism.** Vertical specialization means that production is split into sequential stages across countries. Goods-in-process cross borders multiple times. A tariff is paid at each crossing, so a global tariff cut reduces the cumulative cost of producing vertically specialized goods by a multiple of the tariff cut.
- **Internal margin.** Existing vertically specialized goods become cheaper to produce when tariffs fall, so the volume of trade embedded in these goods expands more than in standard one-stage trade.
- **External margin.** Goods that were previously produced entirely inside one country can become vertically specialized once tariffs fall below a threshold. This discrete switching creates a nonlinear trade response.
- **Quantitative result.** Under benchmark calibrations, the vertical model explains 35.1 percent to 53.0 percent of U.S. manufacturing export growth from 1962 to 1999, compared with 13.3 percent to 29.4 percent in the corresponding one-stage model.
- **Economic takeaway.** Trade costs are compounded along global production chains. Therefore, small tariff reductions can produce large trade responses when production itself is organized as a chain of cross-border tasks.

3 Incremental contribution of the paper

- **Moves from trade in goods to trade in stages.** Standard models treat an internationally traded good as produced in one country and consumed in another. Yi models goods as produced through sequential stages that may be located in different countries.

- **Explains magnification without extreme elasticities.** Rather than assuming a very high elasticity of substitution, the paper shows that the same tariff reduction can have a larger effective cost impact when intermediate goods cross borders repeatedly.
- **Explains nonlinearity.** The external-margin switch into vertical specialization gives the model a threshold effect: tariff cuts can have small effects at high tariff levels and much larger effects after tariffs are low enough for fragmented production to be profitable.
- **Links theory to input-output measurement.** The paper connects the model to empirical measures of imported input content of exports (VS) and exports used in other countries' exports (VS1), building on Hummels, Ishii, and Yi.
- **Provides a calibrated dynamic Ricardian framework.** The paper embeds sequential production into a two-country dynamic Ricardian model with capital, labor, tariffs, and endogenous specialization patterns.
- **Shows the limits of standard models explicitly.** Before introducing the vertical model, the paper demonstrates that Ricardian, monopolistic-competition, and international RBC/Armington models need counterfactually high elasticities to fit trade growth.
- **Reinterprets welfare gains from trade liberalization.** Because vertical specialization expands the scope for specialization across stages, the welfare gain from tariff reductions is larger than in a one-stage model.

4 The empirical puzzle

4.1 Magnitude of trade growth

Let X/Y denote the manufacturing export share of output. The empirical fact motivating the paper is

$$\frac{(X/Y)_{1999}}{(X/Y)_{1962}} \gg 1,$$

while the average manufacturing tariff rate τ falls by only about 11 percentage points. A standard one-stage model typically implies a trade response of the form

$$\frac{d \log(X/Y)}{d \log(1 + \tau)} \approx -\sigma,$$

where σ is an elasticity of substitution or trade elasticity. Matching the data would require σ around 12 or higher, while commonly used and estimated values are closer to two to six.

Economic message. The puzzle is not that tariffs do not matter. The puzzle is that tariffs appear to matter too much relative to what standard one-stage trade mechanisms predict.

4.2 Nonlinear response

The tariff-trade relation is nonlinear. Early tariff reductions generate relatively modest export growth; later tariff reductions generate very large export growth. In the data, the implied elasticity of trade with respect to tariffs is much larger after the mid-1980s than before.

Economic message. A successful model must do more than raise the average elasticity. It must also explain why trade becomes more sensitive to tariffs as tariffs fall.

5 What is vertical specialization?

5.1 Definition

Vertical specialization occurs when three conditions hold:

1. A good is produced in multiple sequential stages.
2. At least two countries contribute value added at different stages.
3. At least one country imports an input, uses it in production, and then exports some of the resulting output.

The paper emphasizes that vertical specialization is not the same as intermediate-goods trade. It is specifically the subset of intermediate-input use that is connected to export production.

5.2 Measures: VS and VS1

The imported-input content of exports is measured as

$$VS_{ki} = \left(\frac{\text{imported intermediates}_{ki}}{\text{gross output}_{ki}} \right) \text{exports}_{ki},$$

where k denotes country and i denotes industry or good.

The second measure captures a country's exports that become imported inputs into other countries' exports:

$$VS1_{ki} = \sum_j \left(\frac{\text{exports}_{ji}}{\text{gross output}_{ji}} \right) \text{exported intermediates}_{kji}.$$

Interpretation. VS looks from the import side of export production: how much foreign input is embodied in my exports? $VS1$ looks from the export side: how much of my exports are used abroad to produce third-country exports?

5.3 Why VS matters quantitatively

The empirical evidence suggests that vertical specialization became a large and growing component of trade. For the United States, vertical specialization accounted for at least about 21.9 percent of exports by 1997, and growth in $VS + VS1$ accounted for roughly 30 percent of U.S. export-share growth between 1962 and 1997.

Economic message. Since vertical specialization itself grew rapidly, a model that ignores it leaves out one of the most important sources of measured trade growth.

6 Why standard models struggle

6.1 Ricardian continuum-goods model

Yi first considers a Dornbusch-Fischer-Samuelson Ricardian model with a continuum of goods. Preferences are

$$U(c) = \int_0^1 \frac{c(z)^v}{v} dz,$$

so the elasticity of substitution across goods is

$$\sigma = \frac{1}{1-v}.$$

Home and foreign unit labor requirements are specified symmetrically:

$$a(z) = 1 + z, \quad a^*(z) = 2 - z.$$

Tariffs create a range of nontraded goods. When tariffs fall, the nontraded range shrinks and trade expands. But to match the observed export-share growth from a 15 percentage point tariff reduction, the model needs σ around 12 or 13.

6.2 Monopolistic competition model

The monopolistic-competition model has increasing returns at the firm level:

$$l_i = a + bx_i,$$

with CES utility over varieties:

$$U = \left(\sum_{i=1}^n c_i^v \right)^{1/v}.$$

The trade elasticity with respect to tariffs is approximately the elasticity of substitution. The paper shows that the model again needs a very high elasticity, around 12 or 13, to match the data.

6.3 International RBC/Armington model

Yi also considers a two-country international RBC model with an Armington aggregator:

$$C_t + I_t = \left[\omega_1 D_t^{1-\alpha} + (1 - \omega_1) X_t^{*1-\alpha} \right]^{1/(1-\alpha)},$$

where $1/\alpha$ is the elasticity of substitution between domestic and imported goods. Once again, matching the observed trade growth requires an elasticity around 14.

6.4 Lesson from the standard models

Across comparative-advantage, increasing-returns, and Armington frameworks, the same problem appears:

- trade grows too little from observed tariff reductions unless elasticities are implausibly large;
- the tariff-trade response is too linear;
- any nonlinearity goes in the wrong direction or is too weak.

Economic message. Standard models treat the value of a good as crossing a border once. Vertical specialization allows the same final good to generate multiple border crossings before it reaches consumers.

7 The dynamic Ricardian model with vertical specialization

7.1 Environment

There are two countries, Home H and Foreign F , and two factors, capital and labor. Each country consumes and invests a nontraded final good. The final good is produced through three stages:

1. Stage 1: a continuum of tradable intermediate inputs is produced using capital and labor.
2. Stage 2: the stage-1 input, capital, and labor are used to produce a second-stage tradable good.
3. Stage 3: the continuum of stage-2 goods is assembled into a nontraded final good.

7.2 Stage 1 production

For country $i \in \{H, F\}$ and good $z \in [0, 1]$,

$$y_1^i(z) = A_1^i(z) k_1^i(z)^\alpha l_1^i(z)^{1-\alpha}.$$

Here $A_1^i(z)$ is country i 's total factor productivity in stage 1 for good z .

7.3 Stage 2 production

Stage 2 uses the stage-1 good as an input:

$$y_2^i(z) = x_1^i(z)^v \left[A_2^i(z) k_2^i(z)^\alpha l_2^i(z)^{1-\alpha} \right]^{1-v}.$$

The parameter v is the cost share or input share of the stage-1 good in stage-2 production. In the benchmark calibration, $v = 2/3$, consistent with the idea that intermediate inputs account for a large part of gross production.

7.4 Stage 3 aggregator

Stage-2 goods are assembled into a nontraded final good using a CES aggregator:

$$Y^i = \left\{ \int_0^1 x_2^i(z)^{(\theta-1)/\theta} dz \right\}^{\theta/(\theta-1)}.$$

In the special case $\theta = 1$, the aggregator becomes Cobb-Douglas:

$$Y^i = \exp \left\{ \int_0^1 \ln x_2^i(z) dz \right\}.$$

7.5 Tariffs and trade costs

An ad valorem tariff τ applies whenever a good crosses the border. If a Home stage-2 firm imports the stage-1 input, its input cost is multiplied by $(1 + \tau)$. If a stage-2 output is exported back across the border, another tariff applies.

Key mechanism. A one-stage traded good pays a tariff once. A vertically specialized good can pay tariffs multiple times. Therefore the effective wedge created by a tariff is larger for vertically specialized production.

8 Production patterns and cutoffs

For each good z , there are four possible production patterns:

- HH : Home produces both stage 1 and stage 2.
- FF : Foreign produces both stage 1 and stage 2.
- HF : Home produces stage 1 and Foreign produces stage 2.
- FH : Foreign produces stage 1 and Home produces stage 2.

The patterns HF and FH are vertically specialized, because the good-in-process crosses the border between stages.

Define relative productivities:

$$A_1(z) = \frac{A_1^H(z)}{A_1^F(z)}, \quad A_2(z) = \frac{A_2^H(z)}{A_2^F(z)}.$$

Vertical specialization occurs when one country has comparative advantage in stage 1 and the other has comparative advantage in stage 2. Under free trade, the HF pattern is efficient when it is cheaper for Home to produce stage 1 and for Foreign to produce stage 2. In words:

stage-1 cost advantage of Home and stage-2 cost advantage of Foreign

are sufficiently strong to overcome the alternative of producing both stages in one country.

8.1 Free-trade cutoff intuition

Figure 5 in the paper shows three regions:

$$[0, z_l] : HH, \quad [z_l, z_h] : HF, \quad [z_h, 1] : FF.$$

The middle region is vertically specialized. The left cutoff separates goods produced entirely in Home from goods whose first stage is in Home and second stage is in Foreign. The right cutoff separates vertically specialized goods from goods produced entirely in Foreign.

8.2 Tariffs squeeze vertical specialization

Tariffs create wedges around free-trade cutoffs. From the perspective of a Home consumer, vertically specialized HF goods can be taxed twice: once when the stage-1 good is exported to Foreign and again when the stage-2 good returns to Home. This squeezes the range of HF goods.

Economic message. Tariffs are especially costly for vertical specialization because they tax the same value-added chain more than once.

9 Derivation of magnification and nonlinearity

9.1 External margin without vertical specialization

Consider the simplified static, one-factor version. If tariffs are high enough that there is no vertical specialization, the cutoff z_l solves

$$q = (1 + \tau)A(z_l),$$

where q is the relative wage and $A(z)$ is overall relative productivity.

Taking log changes and holding relative wages fixed gives

$$\widehat{\left(\frac{z_l}{1 - z_l}\right)} = \left[\frac{z_l}{1 - z_l} \frac{1}{\eta_A} \right] \widehat{(1 + \tau)},$$

where $\eta_A < 0$ is the elasticity of relative productivity with respect to z .

Interpretation. Trade responds to tariff reductions through the standard comparative-advantage cutoff. The smaller the absolute slope of relative productivity, the larger the response.

9.2 External margin with vertical specialization

When vertical specialization is possible, the cutoff condition becomes

$$q = (1 + \tau)^{(1+v)/(1-v)} A_2(z_l).$$

Log differentiating gives

$$\widehat{\left(\frac{z_l}{1 - z_l}\right)} = \left[\frac{1 + v}{1 - v} \right] \left[\frac{z_l}{1 - z_l} \frac{1}{\eta_{A_2}} \right] \widehat{(1 + \tau)}.$$

The term

$$\frac{1 + v}{1 - v}$$

is the external-margin magnification factor. With $v = 2/3$,

$$\frac{1 + v}{1 - v} = \frac{5/3}{1/3} = 5.$$

Interpretation. For a given tariff reduction, the cutoff moves about five times more in the vertical model than in a comparable one-stage model, holding the productivity elasticity fixed.

9.3 External-margin nonlinearity

At high tariffs, no vertical specialization occurs. Trade grows only through the one-stage cutoff. Once tariffs fall below a threshold, vertical specialization becomes profitable for some goods. At that point the relevant cutoff equation changes from the standard expression to the magnified expression.

Economic message. The model predicts weak trade responses at high tariffs and strong trade responses once tariffs become low enough to support cross-border production chains.

9.4 Internal margin

The internal margin concerns goods that are already vertically specialized. Holding the set of vertically specialized goods fixed, lower tariffs still reduce the cost of producing those goods by more than in a standard model.

With two traded stages, the import share response contains an additional term reflecting the tariff paid on imported intermediate inputs. With n traded stages and intermediate-input share v , the reduction in cost per one-percentage-point tariff reduction is proportional to

$$1 + v + v^2 + \dots + v^{n-1} = \frac{1 - v^n}{1 - v}.$$

As n rises, the magnification effect grows.

Interpretation. The more roundabout and fragmented production becomes, the more strongly trade responds to tariff reductions.

10 Calibration

10.1 Period, countries, and target

The simulation begins in 1962 and treats one period as one year. The two countries represent the United States and the rest of the developed world. The model focuses on manufacturing because manufacturing dominates developed-country merchandise trade and vertical specialization is particularly important in manufacturing.

The empirical target is the growth of the U.S. manufacturing export share of manufacturing GDP from 1962 to 1999. Yi adjusts the series for changing relative country size and smooths it for oil-price and exchange-rate movements.

10.2 Tariff series

The tariff series is constructed from U.S., E.C., and Japanese manufacturing tariffs. Yi uses a simple average of the U.S. tariff and a trade-weighted rest-of-world tariff. Tariffs fall sharply after the Kennedy Round, then more gradually after the Tokyo and Uruguay Rounds.

10.3 Main parameter choices

The benchmark choices are:

$$\begin{aligned}\beta &= 0.96, & \alpha &= 0.36, & \delta &= 0.13, \\ v &= 2/3, & \theta &= 1.\end{aligned}$$

The share $v = 2/3$ reflects the fact that value added is roughly one-third of gross manufacturing output, so intermediate inputs account for the remaining two-thirds.

10.4 Relative productivity curves from revealed comparative advantage

The hardest part of calibration is constructing the relative productivity curves $A_1(z)$ and $A_2(z)$. Yi uses Balassa-style revealed comparative advantage (RCA):

$$RCA_{i,j} = \frac{X_{i,j}/X_i}{X_{W,j}/X_W}.$$

The paper extends this idea to production stages:

$$RCA_{i,j,k} = \frac{X_{i,j,k}/X_i}{X_{W,j,k}/X_W},$$

where k denotes stage 1 or stage 2.

Interpretation. Industries are ordered by RCA in their stage-1 counterparts, and quadratic curves are fitted to approximate $A_1(z)$ and $A_2(z)$. This maps observed export patterns into relative stage-specific productivities.

10.5 Narrow and broad benchmark cases

- **Narrow case.** Stage-1 industries include a selected group of intermediate-goods sectors such as paper, industrial chemicals, petroleum products, rubber and plastic, nonmetallic minerals, iron and steel, nonferrous metals, and electrical apparatus.
- **Broad case.** Stage-1 industries include all manufacturing industries. This allows stage-2 industries themselves to supply intermediate inputs to other stage-2 industries.

The broad case creates more scope for vertical specialization because the gap between $A_1(z)$ and $A_2(z)$ is generally larger.

11 Main quantitative results

11.1 Benchmark export growth

The narrow benchmark case explains 74.8 percent export growth compared with 213.0 percent in the adjusted U.S. data from 1962 to 1999. This is 35.1 percent of the data. The broad benchmark case explains 113.0 percent export growth, or 53.0 percent of the data.

Economic message. Vertical specialization does not explain all trade growth, but it substantially improves over a one-stage model and accounts for a large share of the observed increase.

11.2 Nonlinearity

The vertical model generates a stronger trade elasticity when tariffs are lower. In the narrow case, the model's tariff elasticity rises from 3.1 in 1962-1976 to 11.0 in 1976-1999. This is qualitatively consistent with the data, where the elasticity rises from 6.9 to 28.4.

Economic message. The model captures the direction of nonlinearity, although it does not fully match the magnitude of the late-period surge.

11.3 Vertical specialization growth

In the narrow case, $VS + VS1$ rises from 0 percent of GDP in 1962 to 2.83 percent in 1997. In the broad case, it rises to 5.46 percent, close to the U.S. data value of 5.54 percent.

Economic message. The broad model nearly matches the growth of vertical specialization, suggesting that the calibrated stage-productivity structure can reproduce the rise of production fragmentation.

11.4 Comparison with one-stage model

The one-stage model combines the two production stages into one:

$$y_2^i(z) = A^i(z)k^i(z)^\alpha l^i(z)^{1-\alpha},$$

where

$$A(z) = A_1(z)^v A_2(z)^{1-v}.$$

This model removes the possibility of cross-border stage specialization. The one-stage model explains only 13.3 percent of export growth in the narrow case and 29.4 percent in the broad case.

Economic message. The explanatory power comes specifically from allowing production stages to be located in different countries, not merely from Ricardian heterogeneity.

11.5 Welfare gains

Lowering tariffs from about 14 percent to about 3 percent increases steady-state consumption by 3.2 percent in the narrow case and 4.7 percent in the broad case. These gains are larger than in the one-stage model because vertical specialization creates additional specialization opportunities.

12 Robustness and sensitivity

12.1 Alternative RCA calibrations

Yi calibrates the model using textiles and electronics as representative industries. These industries are highly vertically specialized. In the broad calibration, both industries can generate more than 100 percent of observed export growth, showing that some sectors have enough stage-level comparative-advantage differences to produce very strong responses.

12.2 Dynamic transition

The paper also solves a dynamic transition in which the world begins at the 1962 tariff steady state and then follows the observed tariff path. The dynamic export results are similar to the sequence of steady-state results because capital accumulation occurs in both countries and has little effect on relative country size.

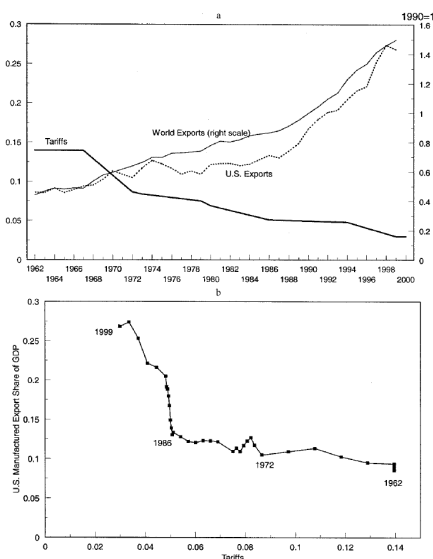
12.3 Parameter variation

Changing the capital share, the input share v , and the elasticity of substitution in the stage-3 aggregator does not overturn the main result. In particular, the model does not rely on extreme substitution elasticities.

Economic message. The central force is the repeated border-crossing structure, not a fragile choice of a single elasticity.

13 Figures and tables

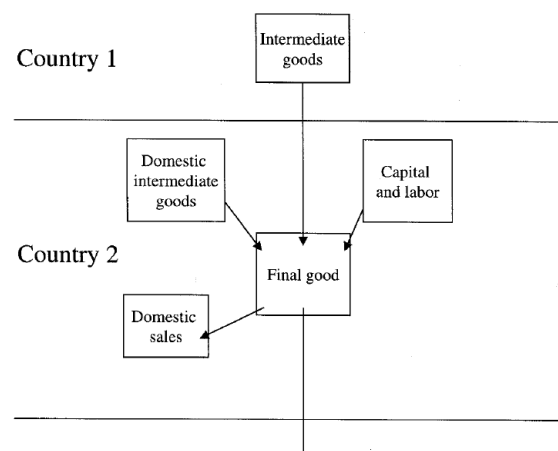
13.1 Figure 1. Manufacturing export shares and tariffs and Figure 2. Example of vertical specialization



(a) Figure 1. Manufacturing export shares and tariffs.

VERTICAL SPECIALIZATION

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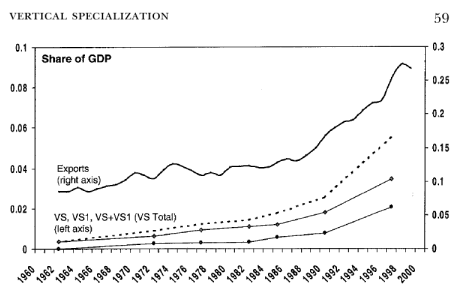


(b) Figure 2. Example of vertical specialization.

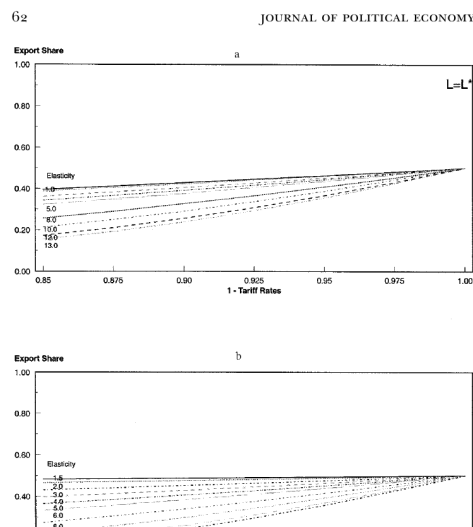
Read. Figure 1 shows the core empirical puzzle: tariff rates decline gradually, while manufacturing exports rise sharply and nonlinearly. Figure 2 shows the conceptual mechanism: a country imports

intermediates, combines them with domestic factors, and exports the resulting good. **Economic message:** the paper shifts attention from trade in final goods to trade generated by cross-border production sequences.

13.2 Figure 3. U.S. vertical specialization and exports and Figure 4. Standard models and tariff responses



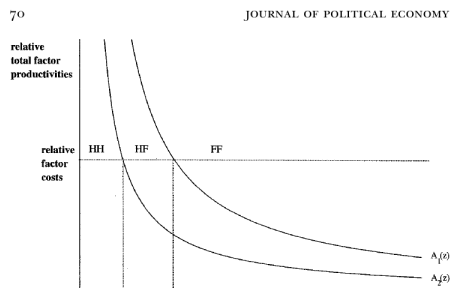
(a) Figure 3. U.S. vertical specialization and exports.



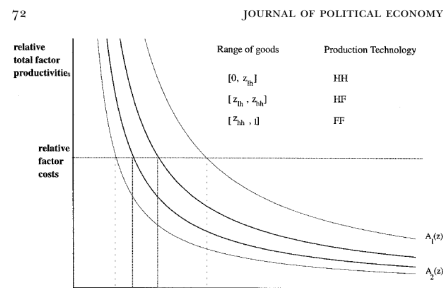
(b) Figure 4. Standard models and tariff responses.

Read. Figure 3 documents the growth of U.S. VS , $VS1$, and total vertical specialization as a share of GDP. Figure 4 shows that standard Ricardian and monopolistic-competition models require very high elasticities to generate large trade responses. **Economic message:** vertical specialization is empirically important and theoretically needed because one-stage models underpredict trade growth.

13.3 Figure 5. Vertical model under free trade and Figure 6. Vertical model with tariffs



(a) Figure 5. Vertical model under free trade.



(b) Figure 6. Vertical model with tariffs.

Read. Figure 5 shows the free-trade cutoff structure: *HH*, *HF*, and *FF* regions. Figure 6 shows how tariffs create wedges and shrink the vertical-specialization region. **Economic message:** tariff reductions expand the range of goods for which cross-border stage specialization is efficient.

13.4 Figure 7. Stage 1 and stage 2 input-output relations and Table 1. Parameters in benchmark calibrations

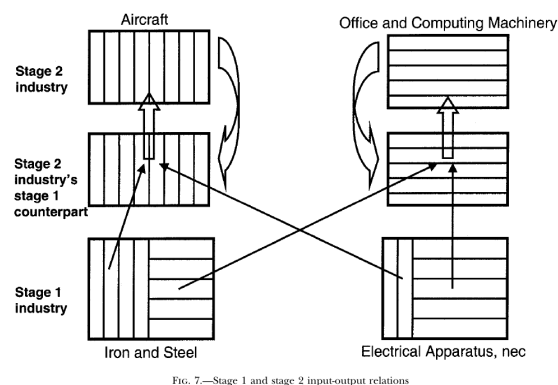


FIG. 7.—Stage 1 and stage 2 input-output relations

(a) Figure 7. Stage 1 and stage 2 input-output relations.

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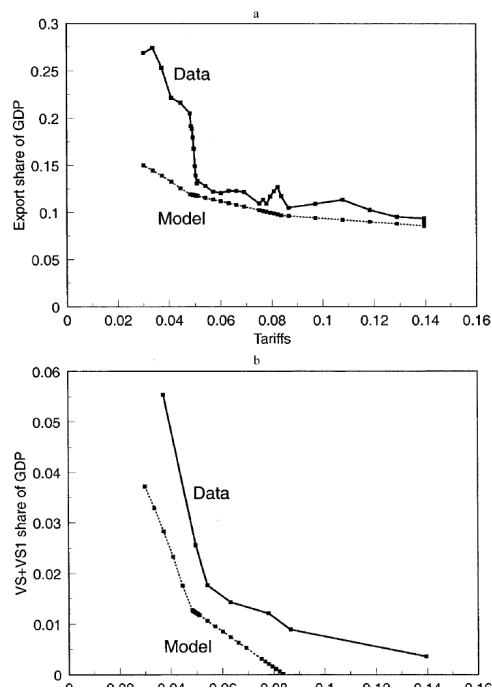
TABLE 1
PARAMETERS USED IN BENCHMARK CALIBRATIONS

Parameter	Parameter Name	Parameter Value
β	Preference discount factor	.96
α	Capital's share in production	.36
δ	Depreciation rate on capital	.13
θ	Share of first-stage output in second-stage production	.67
σ	Elasticity of substitution in stage 3 aggregator	1
$A_1(z)$	Narrow benchmark case stage 1 relative productivity	$1.26z^2 - 2.53z + 1.88$
$A_2(z)$	Narrow benchmark case stage 2 relative	$3.095z^2 - 3.38z + 1.63$

(b) Table 1. Parameters in benchmark calibrations.

Read. Figure 7 explains how the empirical calibration links stage-2 industries to constructed stage-1 counterparts. Table 1 reports the benchmark parameters and the fitted productivity curves. **Economic message:** the calibration translates observed input-output and trade data into stage-specific comparative advantage.

13.5 Figure 8. Narrow case: exports and $VS + VS1$ against tariffs and Table 2. Benchmark vertical-model results



(a) Figure 8. Narrow case: exports and $VS + VS1$ against tariffs.

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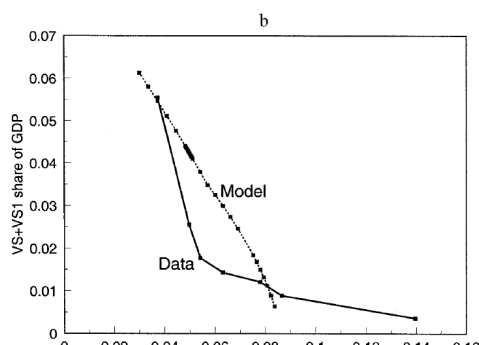
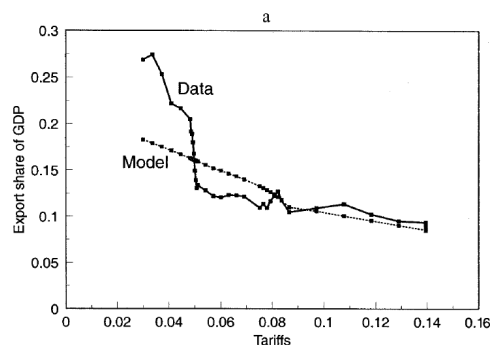
TABLE 2
RESULTS FROM BENCHMARK VERTICAL MODEL

U.S. Data*	Narrow Case	Broad Case
A. Export Growth (Percent)		
1962-99 [†]	213.0	113.0
1962-76	36.2	47.4
1976-99	130.0	44.6
1962-89	73.9	87.2
1989-99	80.1	13.8
B. Elasticity of Export Growth with Respect to Tariffs		
1962-99	22.0	11.8
1962-76	6.9	9.0
1976-99	28.4	9.7
1962-89	9.4	11.1
1989-99	42.1	7.3
C. Export Share of GDP: Root Mean Square Error		
1962-99	.049	.032
1962-76	.015	.006
1976-99	.015	.006

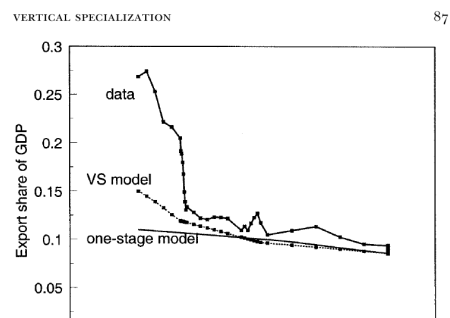
(b) Table 2. Benchmark vertical-model results.

Read. Figure 8 shows that the narrow model matches early trade growth and captures some nonlinearity but underpredicts the later export surge. Table 2 reports that the narrow and broad vertical models explain 35.1 percent and 53.0 percent of overall export growth. **Economic message:** vertical specialization is quantitatively large but not the entire story.

13.6 Figure 9. Broad case: exports and $VS + VS1$ against tariffs and Figure 10. Vertical model versus one-stage model



(a) Figure 9. Broad case: exports and $VS + VS1$ against tariffs.



(b) Figure 10. Vertical model versus one-stage model.

Read. Figure 9 shows that the broad model fits export growth and vertical-specialization growth better than the narrow model. Figure 10 compares the vertical model with the one-stage model in the narrow case. **Economic message:** much of the additional explanatory power comes from allowing goods-in-process to cross borders before final assembly.

13.7 Table 3. Vertical model versus standard model and Table A1. Sources for tariff series

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	NARROW CASE				BROAD CASE	
	U.S. DATA (Adjusted)	Vertical Model	One-Stage Model	Vertical Model	One-Stage Model	
	A. Export Growth (Percent)					
1962-99*	213.0	74.8	28.3	113.0	62.7	
1962-76	36.2	16.2	18.1	47.4	32.9	
1976-99	130.0	50.4	8.7	44.6	22.4	
1962-89	73.9	37.9	24.3	87.2	50.3	
1989-99	80.1	26.8	3.2	13.8	8.2	
B. Elasticity of Export Growth with Respect to Tariffs						
1962-99	22.0	7.8	2.9	11.8	6.5	
1962-76	6.9	3.1	3.4	9.0	6.2	
1976-99	28.4	11.0	1.9	9.7	4.9	
1962-89	9.4	4.8	3.1	11.1	6.4	
1989-99	42.1	14.1	1.7	7.3	4.3	
C. Export Share of GDP: Root Mean Square Error						

(a) Table 3. Vertical model versus standard model.

TABLE A1
SOURCES FOR THE TARIFF SERIES

Year	Source
1962-66	Pre-Kennedy Round; U.N. Conference on Trade and Development (1968, tables A.2, A.3, A.8)*
1967, 1972	U.N. Conference on Trade and Development (1968, tables A.2, A.3, A.8)
1968-71	Kennedy Round phase-in
1973	United States
	Cline et al. (1978, table 2-1), using weights on semifinished manufactures and finished manufactures from U.N. Conference on Trade and Development (1968)
ROW	Interpolation
1974-78	Interpolation
1979, 1986	El-Agraa (1994, table 21.5), using weights on semifinished manufactures and finished manufactures from U.N. Conference on Trade and Development (1968)
1980-85	Tokyo Round phase-in
1987-93	Interpolation
1994, 1999	Schott (1994, table 7) and El-Agraa (1994, table 21.5)
1995-98	Uruguay Round phase-in
2000+	Same as 1999

* Balassa (1965, table 4) reports values about two percentage points less than these values.

(b) Table A1. Sources for tariff series.

Read. Table 3 reports that the vertical model outperforms the one-stage model in export growth, tariff elasticity, and root mean square error. Table A1 documents how the tariff series is assembled from several sources and GATT phase-in schedules. **Economic message:** the quantitative difference between the vertical and one-stage models is not simply due to data construction; it comes from the model's production-chain structure.

13.8 Figure B1. Narrow-case $A_1(z)$ and $A_2(z)$ and Figure B2. Estimated productivity curves

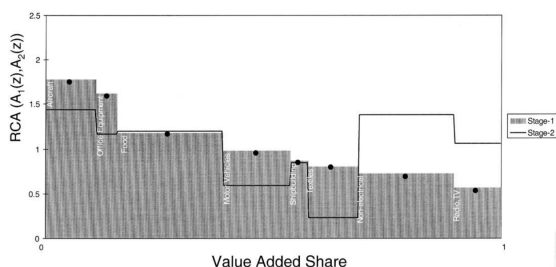
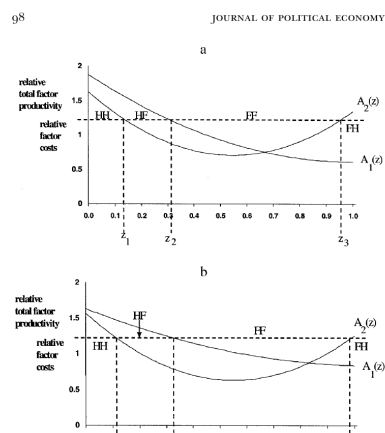


FIG. B1.—Narrow case $A_1(z)$, $A_2(z)$

(a) Figure B1. Narrow-case $A_1(z)$ and $A_2(z)$.



(b) Figure B2. Estimated productivity curves.

Read. Figures B1 and B2 show the empirical construction of the relative productivity curves used in calibration. **Economic message:** the shape and relative position of $A_1(z)$ and $A_2(z)$ determine the potential range of vertical specialization.

13.9 Figure B3. Textiles broad-case estimated $A_1(z)$, $A_2(z)$, and $A(z)$ curves

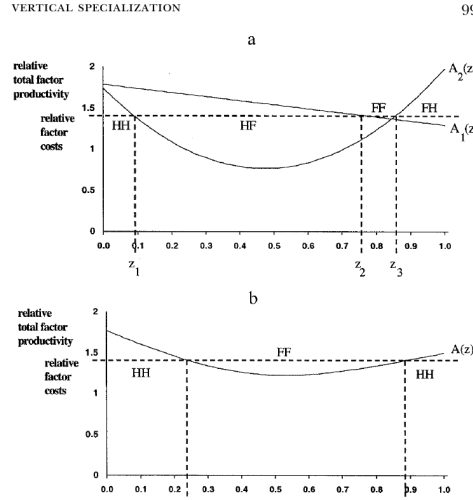


Figure B3. Textiles broad-case estimated $A_1(z)$, $A_2(z)$, and $A(z)$ curves.

Read. Figure B3 illustrates how sector-specific calibration can generate strong scope for vertical specialization. **Economic message:** some sectors have stage-productivity patterns that are especially favorable to production fragmentation.

14 Interpretation and intuition

14.1 Why repeated border crossings matter

Suppose a final good requires several stages and each stage is located in a different country. If the good-in-process crosses borders repeatedly, then the total tax wedge is not just $1 + \tau$ but approximately a product of several terms like $(1 + \tau)$. A tariff reduction therefore lowers the cumulative production cost by a multiple of the direct tariff cut.

This is why a modest tariff reduction can generate large trade growth even when substitution elasticities are moderate.

14.2 Why trade growth can accelerate

At high tariff rates, the cumulative tariff cost makes fragmented production unattractive. As tariffs fall, a threshold is reached where it becomes profitable to split production across countries. Once this threshold is crossed, two things happen:

1. some goods switch into vertical specialization;
2. existing vertically specialized chains expand because their cumulative cost falls.

The threshold creates nonlinearity. Trade can respond weakly at first and then strongly later.

14.3 Why vertical specialization raises welfare gains

A standard model allows countries to specialize in goods. The vertical model allows countries to specialize in *stages within goods*. This creates an additional set of specialization opportunities, so the welfare gain from tariff reductions is larger.

15 Connection to broader literature

- **Ricardian trade.** The model builds on Dornbusch, Fischer, and Samuelson by retaining continuum-goods comparative advantage but introducing sequential production.
- **Global value chains.** The paper is an early quantitative framework for thinking about global value chains and production fragmentation.
- **Input-output trade measurement.** The paper connects model variables to input-output based measures of imported-input content of exports.
- **Trade elasticity interpretation.** Gravity or trade-elasticity estimates may conflate substitution elasticities with production-chain magnification. If vertical specialization is ignored, estimated tariff elasticities may look implausibly high.
- **Welfare from trade.** Because goods can be split across countries at the stage level, gains from trade liberalization can exceed those in one-stage models.

16 Limitations and caveats

- **Two countries and two traded stages.** Real global value chains often involve many countries and many stages. The model may understate the magnification effect because it allows only two traded stages.
- **Tariffs as the main driver.** The paper focuses on tariffs. Other forces - transportation costs, communication costs, containerization, trade reforms in emerging markets, and technology that increases the number of separable stages - may also matter.
- **RCA-to-productivity mapping.** Revealed comparative advantage is an imperfect proxy for relative productivity. The calibration assumes that RCA provides useful information about stage-specific comparative advantage.
- **Incomplete late-period fit.** The model generates nonlinearity, but it does not fully match the size of the export surge after the late 1980s.
- **No endogenous stage creation.** The number of stages is fixed. In reality, technological change may make production more divisible over time.

17 How to use this paper for research ideas

- **Trade shocks and organizational reallocation.** The paper shows that small trade-cost changes can have large effects when production is organized in chains. This logic can be extended to tariffs, supply-chain disruptions, and firm-level sourcing decisions.
- **Networks and amplification.** Vertical specialization is a production network mechanism. A shock to one link can be amplified because goods cross borders and firms multiple times.
- **Nonlinear policy effects.** The external-margin threshold is useful for thinking about why policy shocks may have small effects until an organizational boundary is crossed.
- **Measurement.** The paper motivates using input-output tables, customs data, and firm-level import/export linkages to measure where value added crosses borders.
- **Connection to modern GVC research.** Later work on global value chains, trade in tasks, and firm-to-firm networks can be read as extensions of Yi's central idea: trade grows because production itself becomes internationally fragmented.

18 Conclusion

Yi (2003) explains why the growth of world trade can be much larger and more nonlinear than standard one-stage trade models predict. The key mechanism is vertical specialization: production is split into stages across countries, and goods-in-process cross borders multiple times. Because tariffs are paid at each crossing, tariff reductions reduce cumulative production costs by a multiple of the observed tariff cut. This mechanism generates both magnification and threshold-style nonlinearity.

The calibrated model explains more than half of U.S. manufacturing export growth from 1962 to 1999 in the broad benchmark case and nearly all of the observed growth in vertical specialization. It also performs much better than a standard one-stage Ricardian model. The main lesson is that trade liberalization cannot be evaluated only through final-goods trade. Once production chains cross borders repeatedly, small changes in trade costs can reorganize production and generate large measured increases in trade.